**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Evaluating Technology Impacts

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is a key feature of an open-source licence?

2. How does the OS support multitasking?

A) Adds a CPU per app
B) Closes every other app
C) Gives each app all RAM
D) Shares CPU time

3. "A longer recording automatically has finer amplitude accuracy." Correct this. Explain why.

4. A mono clip uses 1,000 samples/s for 3 s at 8 bits/sample. Calculate its raw size in bytes, excluding headers and compression.

## Today's objectives

Explain benefits and drawbacks for different stakeholders. Write balanced six-mark evaluations using specific evidence.

## Key vocabulary - Read, cover, recall

**Evaluate**
Weigh strengths and weaknesses and reach a supported judgement.

**Stakeholder**
Person or group affected by a decision.

**Justification**
Reasons and evidence supporting a choice.

**Privacy**
Control and protection of information about individuals.

**Write one key term and its meaning in your own words:**

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| **Recall previous learning** | Read new knowledge | Practise with guidance | Work independently | Check my understanding |

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Evaluation is not a list** | Describe the proposed technology and its context first. A strong evaluation considers different viewpoints, consequences and realistic trade-offs. |
| **Think about stakeholders** | Users, employees, managers and the wider community may be affected differently. Give real impacts rather than simply saying technology is good or bad. |
| **Consider multiple dimensions** | Discuss appropriate ethical, legal, environmental, cultural or privacy aspects. Do not force every category into every answer if it is irrelevant. |
| **Reach a clear judgement** | Make a reasoned decision that recognises the most important benefit and risk in the given scenario. Explain conditions that would make the decision more acceptable. |

## I Do - One worked example

**A school plans to use automated facial recognition to record attendance. Evaluate TWO benefits and TWO concerns, then give a supported judgement. [6 marks]**

1. Identify the aim: recording attendance quickly and accurately.

2. Automation may reduce administrative time [1] so staff can focus on teaching [1].

3. Face data is sensitive personal information and may produce privacy or fairness concerns [1]; inaccurate recognition could affect pupils [1].

4. A justified judgement [2]: only consider use with lawful safeguards, accuracy checks and a fair non-biometric alternative. Credit reasoned alternative conclusions.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A town council plans to replace most paper communications with a mobile app. Evaluate benefits and drawbacks, with a judgement. [6 marks]

2. A company allows staff to work remotely using cloud services. Evaluate an advantage and a concern. [4 marks]

3. A school installs more computers rather than repairing current ones. Evaluate an environmental and social consequence. [4 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Predict the advice when a service is online-only.**

```
onlineOnly = true
if onlineOnly then
print("Offer alternatives")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

## You Do - Independent exam questions

Q2. State TWO groups who could be affected by a new school app. [2 marks]

Q3. A sports club moves registration entirely online. Explain one benefit and one drawback for members. [4 marks]

Q4. Evaluate a company's proposal to monitor employee locations with a work app. Consider ethical and privacy effects and reach a judgement. [6 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A strong evaluation identifies relevant benefits and harms and supports its conclusion. ☐ T ☐ F
2. Describe the proposed technology and its context first ☐ T ☐ F
3. Users, employees, managers and the wider community may be affected differently ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** It permits access to source code and modification or redistribution under its conditions. ☐

**Q2** D - Shares CPU time ☐

**Q3** It doubles size but does not by itself improve sampling quality. ☐

**Q4** (a) 1,000 x 3 x 8 = 24,000 bits; divide by 8 = 3,000 bytes. ☐

## Code First - Model

```
onlineOnly = true
if onlineOnly then
print("Offer alternatives")
endif
```

**Output: Offer alternatives**

## We Do - Check the model

1. An app may speed communication [1] and reduce paper/postage [1]. People without smartphones/digital skills could be excluded [1], harming fair access [1]. A reasoned judgement [2] could support the app if telephone and paper alternatives remain available. ☐

2. Staff can access work from different locations [1], improving flexibility [1]. Cloud access depends on connection/security safeguards [1], so outages or poor access control could disrupt work or expose data [1]. ☐

3. More computers may improve access for students [1], reducing some classroom waiting or sharing [1]. Replacing working machines creates e-waste/material demand [1], increasing environmental impact [1]. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Pupils [1] and parents/carers [1]. Teachers/administrators also acceptable. ☐

**Q3** Members with devices can register remotely [1], saving travel/time [1]. Members without connectivity or digital skills [1] may be unable to register easily [1]. ☐

**Q4** Location monitoring may help coordinate work or safety [1], giving managers useful scheduling information [1]. It may intrude into employees' privacy [1] or harm trust [1]. A justified conclusion [2] should weigh necessity, lawfulness, transparency, limits and proportionate alternatives. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**